Speaker Information

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Talk Information

Live tissue clearing with SeeDB-Live

Tissue clearing has been widely used for fluorescence imaging of fixed tissues, but not for live tissues due to its toxicity. Here we develop minimally invasive optical clearing media for fluorescence imaging of live mammalian tissues. Light scattering is minimized by adding spherical polymers with low osmolarity to the extracellular medium. A clearing medium containing bovine serum albumin (SeeDB-Live) is minimally invasive to live cells, allowing for structural and functional imaging of live tissues, such as spheroids, organoids, acute brain slices, and the mouse brain in vivo. SeeDB-Live minimally affects the electrophysiological properties and sensory responses of neurons. SeeDB-Live facilitates fluorescence imaging of deep cortical layers in live animals without noticeable toxicity to neurons and animal behavior. We also demonstrate its utility for epifluorescence voltage imaging in acute brain slices and in live animals in vivo. Thus, SeeDB-Live expands the scale and modalities of fluorescence imaging of live mammalian tissues.

Research Overview (for collaboration discussion)

[Current Research Projects]

- Chemical senses (Olfaction and gut-brain axis)
- Neural development in the olfactory bulb and the cerebral cortex
- Synaptic development and neuropsychiatric diseases
- Tissue clearing and light microscopy-based connectomics

[Techniques]

Tissue clearing, multicolor imaging, light microscopy, ex vivo / in vivo 2-photon imaging

Short Bio

I obtained Ph.D. in 2006 at the University of Tokyo under the supervision of Prof. Hitoshi Sakano. After spending four years as a postdoc in the same lab, I started my own lab at RIKEN Center for Developmental Biology in 2010. I then joined Kyushu University as a full professor.